

A CG-Corrected Residual Certificate for Learned Elliptic Solvers

certno

Research Program on Error-Certified Neural Operators

<https://github.com/emmettv1191/certno-phd-research>

Abstract—We propose a CG-corrected a posteriori residual certificate for the discrete elliptic system $A_h u_h = f$. Starting from the standard bound $\|u_h - \hat{u}\|_2 \leq \frac{\|r\|_2}{\lambda_{\min}(A_h)}$, we compute $z \approx A_h^{-1}r$ via CG and form $B_{CG} = \|z\|_2 + \frac{\|r - A_h z\|_2}{\alpha}$ with $\alpha \leq \lambda_{\min}(A_h)$. Across coefficient contrasts, ID/OOD, and real FNO predictions, B_{CG} is near-sharp (effectivity ≈ 1.0) while preserving coverage 1.0 and FNR 0.0. The certificate enables selective verification by accepting predictions when $B_{CG} \leq \tau$.

REPRODUCIBILITY

Code: <https://github.com/emmettv1191/certno-phd-research>. See BENCHMARK_FREEZE.md, THEOREM_AUDIT.md, RESULTS.md.

I. INTRODUCTION

Error certification for learned PDE surrogates is needed to decide when a prediction is trustworthy. For SPD elliptic systems, residual-based bounds provide safety but can be overly conservative with global coercivity constants.

II. THEORY

Let $A_h \succ 0$, $u_h = A_h^{-1}f$, $r = f - A_h \hat{u}$, $e = u_h - \hat{u} = A_h^{-1}r$. For any z , $e = z + A_h^{-1}(r - A_h z)$. Thus

$$\|e\|_2 \leq \|z\|_2 + \frac{\|r - A_h z\|_2}{\lambda_{\min}(A_h)} \leq \|z\|_2 + \frac{\|r - A_h z\|_2}{\alpha}, \quad \alpha \leq \lambda_{\min}(A_h).$$

We take z as CG approximation of $A_h^{-1}r$.

III. ASSUMPTIONS

- (1) Discrete SPD system from coercive $-\nabla \cdot (a \nabla u) = f$, $a \geq a_{\min} > 0$.
- (2) $\alpha = a_{\min} \lambda_{\min}(-\Delta_h) \leq \lambda_{\min}(A_h)$.
- (3) Double-precision arithmetic; CG stopping controls $\|r - A_h z\|$.
- (4) Certification is w.r.t. discrete u_h .

IV. EXPERIMENTS

Table ?? shows comparison across contrasts. With real FNO predictions, coverage/FNR remain 0/1.0. Selective verification accepts only when $B_{CG} \leq \tau$.

TABLE I
ANALYTIC VS CG-CORRECTED (ABSOLUTE L2). EFFECTIVITY $B/\|e\|$.

Contrast	Cov (both)	FNR (both)	Med eff (analytic)	Med eff (CG)
1.5	1.0	0.0	312	1.00
3.0	1.0	0.0	500	1.00
5.0	1.0	0.0	768	1.00
10.0	1.0	0.0	1361	1.00

V. CONCLUSION

The CG-corrected residual certificate is rigorous and near-sharp, enabling safe selective verification of learned elliptic solvers.